Client Report: Passive Pre-Harvest SLF Trap for NYS Industrial Vineyards

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Client(s): Cornell CALS Extension / E&J Gallo Winery / National Grape

Context and Problem Statement

New York State industrial vineyards need a low-labor method to reduce spotted lanternfly (SLF) contamination before machine harvesting. Because many grapes are mechanically harvested as juice or must, SLFs that enter the harvesting stream are difficult to remove without yield loss, quality degradation, or added labor. Our team focused on the pre-harvest sub-problem: attracting and capturing SLFs near vines before they enter the harvester.

The client does not need a full SLF eradication system; they need a passive, vineyard-compatible device that can be placed near edges or hotspots, serviced quickly, and moved as pest pressure changes. Therefore, our design prioritizes passive operation, modular maintenance, low cost, and compatibility with vineyard perimeter deployment.

Final Prototype and Application

Our final prototype is a passive box-shaped trap using methyl salicylate as the lure. Methyl salicylate is held in a removable vial. Cotton balls contact the liquid and act as a passive wick, directing scent toward a funnel. SLFs following the scent enter through the funnel and are immobilized on replaceable sticky panels inside the trap. A removable side panel allows workers to replace sticky sheets and refill the lure without disassembling the full device. An iris-style mechanism adjusts the funnel opening to control entry size.

Use procedure:: remove the side panel, slide out the vial holder, refill methyl salicylate, insert cotton balls, install the sticky sheet in a U-shape inside the trap, close the panel, adjust the iris opening, and place the device near vineyard edges or SLF hotspots. Sticky sheets and lure should be replaced approximately every two weeks, pending field validation.

- **Box Assembly**
  - Press fit the left, right, and back plates into the base plate.
  - Make sure the smooth edges on the left and right panels face the removable side.
  - Align the slots, push the top plate on, and apply glue at the connections.
- **Mount Assembly**
  - Glue one M4 screw into the hole on each mount. Repeat for all four mounts.
  - Orient the mounts so the screws aim outward from the box.
  - Glue the mount extrusions into the small slots on the top plate.
  - Hang the removable side panel on the screws.
- **Funnel Installation**
  - Screw the funnel from above the top plate using two M4 screws and bolts.
- **Chemical Lure Assembly**
  - Poke a hole in the top of the vial using a soldering iron.
  - Roll a cotton ball into a rope and jam one end into the hole in the lid.
  - Fill the vial with approximately 5 mL of methyl salicylate solution.
  - Close the lid, ensuring the cotton rope is in contact with the chemical.
  - Screw the diffuser connector from below the top plate using M3 screws and bolts.
  - Slide the vial into the mounting bracket and screw the bracket and vial into the box.

- **Iris Assembly**
  - Place the blades on the bottom plate, oriented so the iris is at maximum opening.
  - Put on the top plate and adjust the opening to the desired size.
  - Screw the iris from above the top plate using M4 screws and bolts.
- **Knob Assembly**
  - Place the small gear concentric to the large hole on the top plate and make sure it meshes with the large outer gear of the iris.
  - From the top, press fit the knob into the gear.
  - Verify that the knob properly changes the aperture of the iris.
- **Sticky Trap Installation**
  - Cut insect traps into 30 cm by 10 cm rectangles.
  - Cover the non-sticky side with two parallel strips of double-sided tape.
  - Fix multiple stacked layers of 30 cm by 10 cm sticky traps onto the inside of the box.

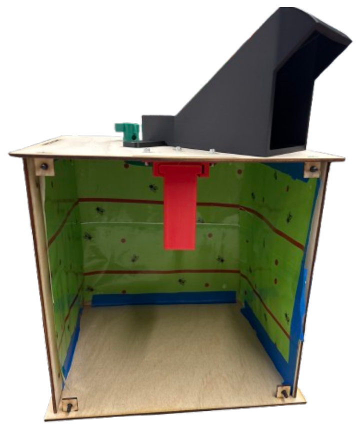

Figure 1: Passive pre-harvest SLF trap prototype with methyl salicylate lure, funnel entry, adjustable iris, and removable maintenance panel.

Success Criteria and Testing Results

Based on client feedback, our revised success criteria focus on measurable performance rather than basic product functions. The trap should not merely open, close, and hold lure; it should meet quantitative thresholds that justify field testing and eventual vineyard deployment.

Because live SLF availability was limited during the prototype period, our testing focused on bench validation of the highest-risk mechanisms: the iris opening, vial-holder integration, and removable panel. The iris opened to 45 mm, which is larger than typical adult SLF dimensions, and it operated repeatedly without functional failure. The vial holder slid in and out without interfering with the iris or funnel, confirming that the lure module can be serviced independently. The removable side panel provided access to the sticky trap area and could be reattached successfully, although sealing and print tolerances should be improved before outdoor use.

These tests confirm that the prototype is mechanically ready for controlled field testing. They do not yet prove biological capture performance. The next test should compare three groups: trap with methyl salicylate, identical trap without methyl salicylate, and standard sticky card control. Decision metrics should include SLFs captured per trap per day, non-target insects captured, lure mass loss, rain resistance, and maintenance time per trap.

Category	Success Criterion	Prototype Result / Next Step
SLF Entry	Iris must open to at least 40 mm, exceeding adult SLF body size.	Prototype opened to 45 mm; entry size is sufficient.
Maintenance	Sticky panel and lure replacement should take under 3 min per trap with no special tools.	Service process took between 45 to 90 seconds across group members; target should be timed formally in field trials.
Durability	Iris and removable panel should survive 100 open-close cycles without fracture or jamming.	500 Repeated operations showed no major failure; printed-part friction should be reduced with lubricants for better performance however.
Lure Life	Methyl salicylate reservoir should last at least 14 days between refills.	Lure in cotton balls lasted over 20 days indoors; evaporation rate must be measured outdoors.
Capture Value	Trap should capture at least 25% more SLFs than a control sticky card over the same period.	Not yet field-tested; this is the key next validation metric.

Table 1: Compact success criteria and current prototype status.

Prototype Details

The iris mechanism provides an adjustable opening at the funnel entrance. A fixed opening would be simpler, but the iris allows users to tune the opening based on site conditions, observed insect size, and non-target insect concerns. The methyl salicylate system is passive rather than powered, avoiding pumps, electronics, or frequent adjustment. Cotton balls increase exposed surface area and guide scent from the vial toward the funnel. The removable side panel is the primary maintenance feature because sticky inserts must be replaced after capture.

The current prototype is best understood as a functional proof-of-concept, not a final vineyard product. A field-ready version should use UV- and rain-resistant materials, improved panel sealing, smoother iris surfaces, a standardized lure cartridge, and a more guided sticky-panel insert.

Conclusion and Recommendation

We recommend advancing this design to controlled field testing rather than immediate vineyard-scale deployment. The prototype demonstrates the correct mechanical functions for a passive pre-harvest trap: lure storage, scent diffusion, adjustable entry, sticky capture, and quick maintenance access. These functions directly address the client's need for low-labor SLF reduction before machine harvesting.

The most important unanswered question is whether the methyl salicylate trap captures meaningfully more SLFs than simpler control traps in actual vineyard conditions. If field testing shows at least 25% higher SLF capture than a control sticky card while maintaining service time under 3 minutes per trap every two weeks, the design is promising for further development. If not, the design should be simplified or redesigned around a stronger lure and capture strategy.

References

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Bill of Materials

Item	Qty.	Est. Unit Cost	Purpose
Laser cut body panels	1 set	\$2–\$4	Structure
3D-printed funnel, iris, side panel, vial holder	1 set	\$8–\$15	Structure
Methyl salicylate solution	1 refill	\$2–\$5	Lure
Small vial	1	\$0.50–\$1.50	Reservoir
Cotton balls	2/refill	<\$0.10	Wick/diffuser
Sticky trap sheet	1/service	\$0.50–\$2.00	Capture surface
Small screws/fasteners	4–8	\$0.50–\$1.00	Assembly
Gasket/tape/adhesive	As needed	\$0.50–\$2.00	Sealing
Mounting hardware/zip ties	As needed	\$0.50–\$2.00	Deployment
Estimated prototype cost		**\$14–\$30**	Per trap

Table 2: Estimated bill of materials for one prototype trap.